Appendix A. Trusted Agent Ecosystem Architecture for Major Events

A.1 Introduction

Major international events such as the FIFA World Cup, Olympic Games, and Paralympic Games are increasingly dependent on large-scale digital ecosystems that span national borders, organizational boundaries, and physical infrastructures. These events bring together governments, border agencies, transportation providers, sporting federations, venue operators, broadcasters, sponsors, public safety organizations, athletes, and millions of participants within a shared operational environment.

As AI capabilities mature, autonomous and semi-autonomous agents are expected to perform an increasing proportion of operational, commercial, and participant-facing activities. These activities range from travel planning, visa processing, credential management, and transportation coordination through to venue operations, athlete performance services, broadcasting, sponsorship activation, and public safety management. The resulting ecosystem creates a new trust challenge: organizations must be able to determine whether a digital agent possesses a valid identity, whether authority has been legitimately delegated, whether the agent remains trustworthy at runtime, and who is accountable for its actions.

This appendix proposes a Trusted Agent Ecosystem Architecture for Major Events as a reference model for understanding these interactions and identifying the trust capabilities required to support them.

A.2 Architectural Overview

The architecture treats a major event as a temporary digital nation composed of multiple interacting trust domains. Rather than viewing travel, ticketing, broadcasting, sponsorship, athlete services, and security as independent systems, the architecture considers them components of a single ecosystem connected through trusted digital agents.

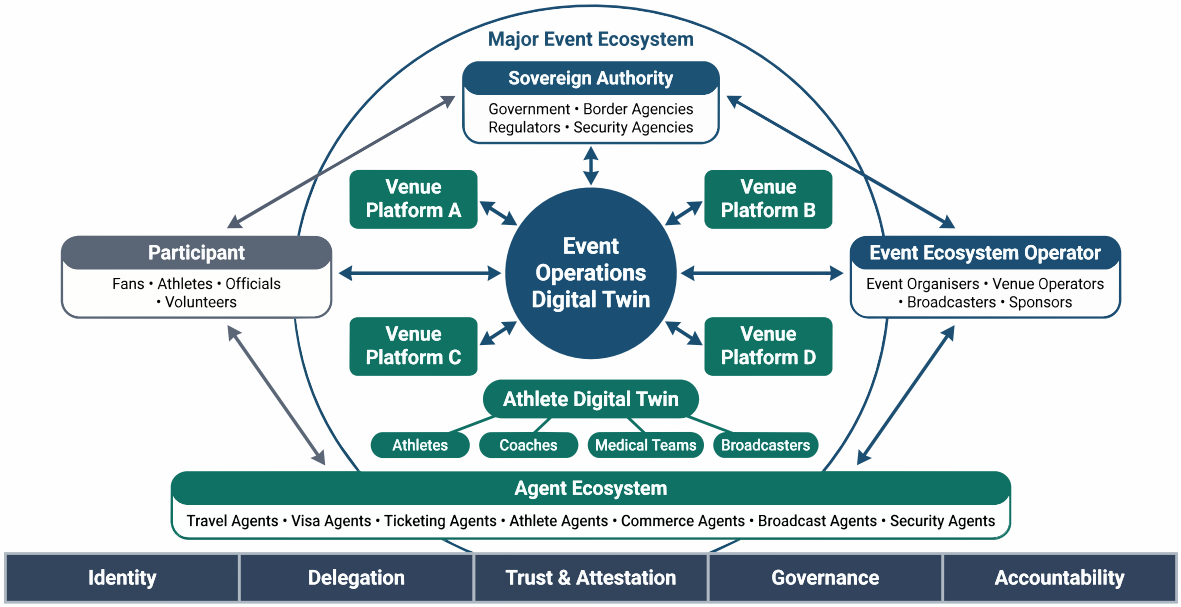
At the centre of the architecture is the **Event Operations Digital Twin**, which provides a continuously updated representation of the event environment and supports operational coordination across organizations and jurisdictions. The digital twin receives information from transportation networks, venues, security systems, broadcasters, sponsors, athlete services, and public authorities, enabling ecosystem-wide situational awareness and decision support.

The architecture is organized around four primary perspectives:

- **Participants**, including fans, athletes, volunteers, officials, and media representatives.
- **Agent Ecosystems**, comprising travel, visa, ticketing, athlete, commerce, broadcasting, sponsorship, and security agents.
- **Event Ecosystem Operators**, including event organizers, venue operators, broadcasters, sporting federations, and commercial partners.
- **Sovereign Authorities**, including governments, regulators, border agencies, and public safety organizations.

Each participant, organization, and infrastructure component may be represented by one or more agents acting under delegated authority. The challenge for every relying party is determining whether that authority is valid, appropriate, and trustworthy.

Figure A1. Trusted Agent Ecosystem Architecture for Major Events



Illustrates the overall architecture, including Participants, Agent Ecosystem, Event Ecosystem Operators, Sovereign Authorities, Event Operations Digital Twin, Venue Platforms, Athlete Digital Twins, and foundational trust capabilities.

A.3 Digital Twins and Platform Ecosystems

A distinguishing feature of the proposed architecture is the integration of multiple digital twins operating within a common trust framework.

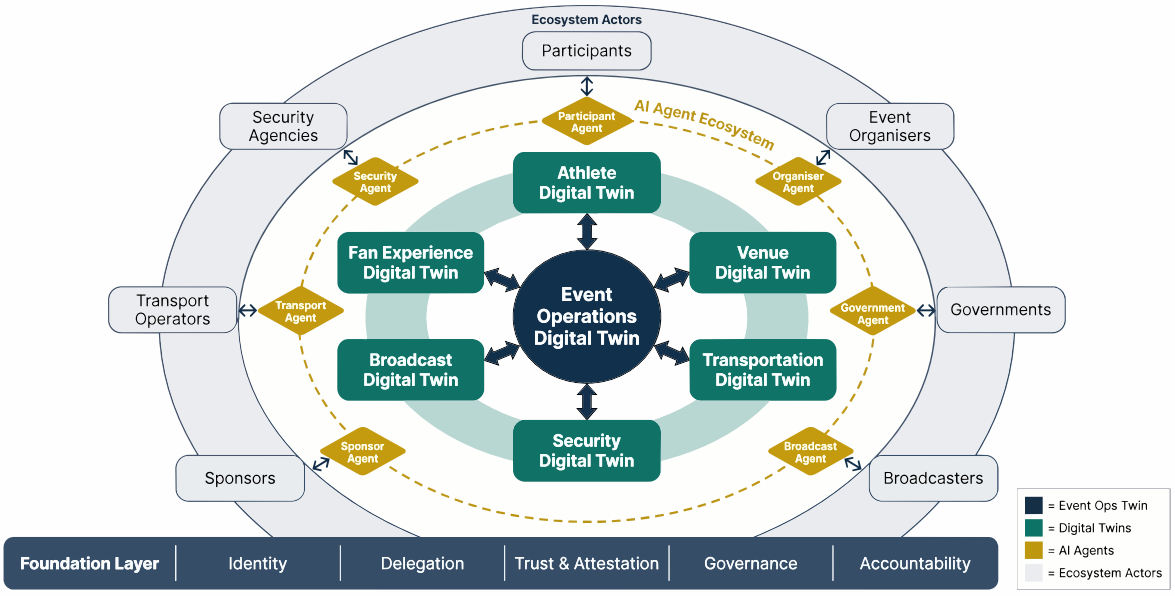
The **Event Operations Digital Twin** acts as the ecosystem coordination layer. It integrates operational information from transportation systems, border operations, venue platforms, public safety services, broadcasters, sponsors, and sporting organizations to support planning, optimization, simulation, and operational decision-making. Decisions generated from the twin may influence mobility, crowd management, security operations, broadcasting activities, and resource allocation.

In parallel, **Athlete Digital Twins** provide trusted representations of athlete health, training, performance, competition readiness, and recovery. These digital representations may be accessed by authorized athletes, coaches, medical teams, sporting federations, and broadcasters. Given the sensitivity and commercial value of athlete data, strong governance mechanisms are required for consent management, privacy protection, provenance verification, authorization, and accountability.

Within the broader event ecosystem, individual venues and stadiums operate as **platform ecosystems**. Venues are no longer merely physical locations; they serve as trusted interaction environments where people, agents, organizations, infrastructure, and digital services converge. Venue platforms facilitate ticketing, accreditation, venue access, commerce, broadcasting, sponsorship activations, crowd management, and mobility services. They represent the primary operational environment in which trust decisions are exercised and enforced.

Together, the Event Operations Digital Twin, Athlete Digital Twins, and Venue Platforms form an integrated digital ecosystem that coordinates the movement of people, information, authority, and value throughout the event lifecycle.

Figure A2. Integrated Digital Twin Ecosystem for Major Events



Illustrates the relationships between the Event Operations Digital Twin, Athlete Digital Twins, Venue Platforms, transportation systems, security systems, broadcasters, sponsors, and the AI agents that coordinate interactions between them.

A.4 Trust Across the Participant Journey

The architecture can also be viewed from the perspective of an individual participant.

A typical fan or athlete journey may begin with ticket purchase and travel planning, followed by visa processing, border clearance, transportation, venue access, event participation, venue commerce, broadcast engagement, and eventual return travel. At each stage, agents increasingly act on behalf of participants, organizations, and public authorities.

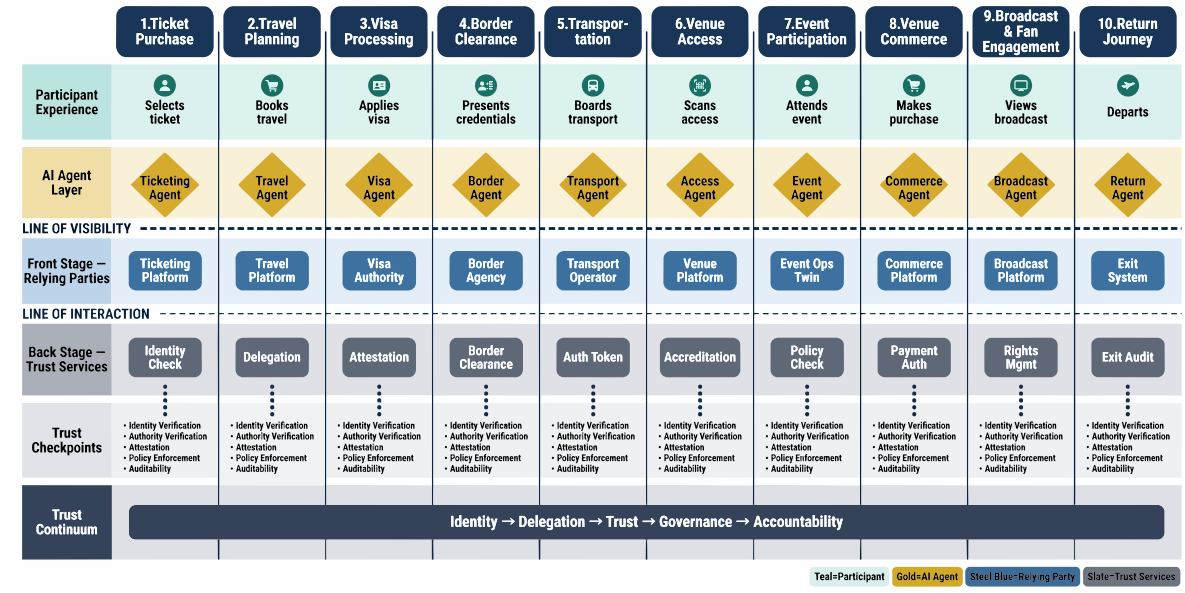
For example, a travel agent may coordinate flights and accommodation, a visa agent may assist with travel authorization, a mobility agent may provide transport services, and a venue agent may facilitate entry and access rights. Broadcasters and sponsors may also interact with participants through personalized digital services and AI-enabled engagement platforms.

Although these interactions differ operationally, they all involve the same fundamental trust question:

> Can the relying party trust that the agent possesses the identity, authority, permissions, and accountability required to perform the requested action?

This journey perspective highlights why trust must be treated as a continuous capability rather than a one-time verification event. Identity, delegated authority, attestation, governance, and accountability must be maintained throughout the lifecycle of every interaction.

Figure A3. End-to-End Major Event Trust Journey



Illustrates the participant journey from ticket purchase through travel, border processing, transportation, venue access, event participation, commerce, broadcasting, and return travel, highlighting trust decisions at each stage.

A.5 Mapping to FG-TIDA Themes

The proposed architecture exercises all major FG-TIDA themes and demonstrates how identity, authority, trust, governance, and accountability interact within a single ecosystem-scale use case.

Table A1. Mapping Major Event Ecosystem Components to FG-TIDA Themes

Architecture Component	Primary Trust Question	Relevant FG-TIDA Themes
Participant	Who may act on my behalf?	Dynamic Identity, Delegation
Agent Ecosystem	Can I prove my authority and trustworthiness?	Delegation, Continuous Trust and Attestation
Venue Platform	Should this requested action be permitted?	Runtime Enforcement, Dynamic Identity
Athlete Digital Twin	Who may access, use, and share athlete information?	Delegation, Dynamic Identity, Privacy
Event Operations Digital Twin	Can operational recommendations be trusted?	Continuous Trust and Attestation, Runtime Enforcement
Event Ecosystem Operator	Should I trust this agent?	Discovery and Cross-Border Trust, Accountability
Sovereign Authority	Is this action lawful and auditable?	Governance, Accountability, Legal Binding

A.6 Conclusion

Major international events provide a uniquely valuable environment for exploring trusted digital agents at ecosystem scale. They combine multiple trust domains, multiple jurisdictions, multiple platform ecosystems, and multiple categories of digital twins within a single operational environment. At the same time, they involve the movement of people, information, authority, and value across organizational and national boundaries.

The Trusted Agent Ecosystem Architecture demonstrates how digital agents, platform ecosystems, and digital twins can be coordinated through a common foundation of identity, delegation, trust, governance, and accountability. As such, major events represent a compelling flagship use case for Trusted Identity for Digital Agents (TIDA) and provide a realistic testbed for future trusted agent ecosystems operating at city, national, and global scale.